

v1.0Deep Dive_ Tau Neutrino Universe

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Tau-Neutrino Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Deep Dive_ Tau Neutrino Universe V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
IceCube ν_τ detection	No tau neutrino observations	IceCube 2023: 10 tau neutrino events from astrophysical oscillation. Validates 3-flavor mixing at PeV energies	HIGH	First observational confirmation of tau neutrino beyond accelerator — major discovery
Mass hierarchy preference	Unknown in V1.0	Normal Hierarchy preferred at 2.7σ (NOvA 2023) + 3.5σ (IceCube 2023). V2.0 defaults to NH: $m_1 < m_2 < m_3$	MEDIUM	V1.0 treats both hierarchies equally; NH now favored
DOUBLE_BANG topology	Not modeled	IceCube double-bang: 50-100m between tau production and decay. V2.0: two connected light cones with decay length $c\tau(E/m_\tau)$	HIGH	Tau neutrino signature entirely absent from V1.0 simulation

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
KATRIN mass bound	< 1.1 eV (older)	KATRIN 2024: $m_\nu < 0.45$ eV — 2.4× tighter. All neutrino masses < 0.45 eV	MEDIUM	Neutrino mass parameter upper bound needs update

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** DOUBLE_BANG, IceCube
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Deep Dive_ Tau Neutrino Universe V1.0 archived | V2.0 is the active specification as of June 2026